

NISHANTH VEDURUVADA

AI Platform Architect

Agentic Platforms • AI Governance • Enterprise AI Modernization • Zero Trust Architecture

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PROFESSIONAL SUMMARY

AI Platform Architect with 9+ years across software engineering, cloud and solution architecture, and application security — now designing the platform and governance layer that lets autonomous systems run inside regulated enterprises. Currently AI Engineer Advisor at NTT DATA, leading architecture and technical governance for 8–10 enterprise AI modernization programs across the Syntphony product ecosystem, directing a team of AI engineers and data scientists. Previously sole architect of the AI Trust Fabric — a NIST 800-207 Zero Trust and AI governance architecture for a national-scale insurance data ecosystem under insurance-regulator oversight, registered as an industrial design with the Government of India. Depth across agent topology and tool protocols (MCP, A2A), enterprise retrieval, identity and authorization, audit and observability, and the cloud platform underneath — in insurance, telecommunications and energy.

SELECTED ACHIEVEMENTS

- Reference architecture for a reusable agentic platform.** Replaced per-engagement rebuilds of the same plumbing with a platform layer: MCP for tool integration (registry, schema discovery, auth, rate limiting, audit) and A2A for cross-domain agent collaboration (registry, router, orchestration, human-in-the-loop approval). Domain agents are built as plugins and inherit identity, audit and approval controls as platform behaviour rather than re-implementing them.
- Registered industrial design, Government of India.** Sole architect of the AI Trust Fabric — NIST 800-207 Zero Trust extended with continuous adaptive trust scoring and an AI agent governance layer, for a national-scale insurance data ecosystem. Co-registered as Design No. 485611-001.
- Peer-reviewed publication on AI agent security.** Co-author, *MCP Guardian* — middleware adding authentication, per-token rate limiting and WAF-style tool-call scanning to the Model Context Protocol without modifying individual MCP servers, evaluated at roughly 3–4 ms median latency overhead.
- Zero Trust migration across 5,000+ users.** Led network segmentation and a SASE rollout replacing perimeter-based access with identity-centric controls — the security grounding the later AI governance architecture is built on.

CORE COMPETENCIES

Enterprise & Solution Architecture • AI Platform Architecture • Agentic & Multi-Agent Systems • Retrieval-Augmented Generation (RAG) • AI Governance • Zero Trust Architecture • Cloud Architecture • Identity & Authorization • Technical Leadership

PROFESSIONAL EXPERIENCE

NTT DATA

Dec 2025 – Present

AI Engineer Advisor

Architecture and technical governance for AI modernization across the Syntphony product ecosystem — Syntphony AI, Open Telecom Networks, Insurance Cloud Migration, Stations, and Commerce & Payments.

- Agentic Platform Architecture:** Engagement teams were independently rebuilding tool integration, agent-to-agent messaging, audit and rate limiting — duplicated effort, inconsistent security posture. Defined a reusable reference architecture standardising MCP as the tool primitive and A2A for cross-domain requests, with domain agents (OSS assurance, inventory, fulfilment, RAN automation) composed as plugins on a shared platform.
- Agent Identity & Governance Controls:** Established per-agent OAuth2 workload identity in place of borrowed user credentials, action-scope enforcement restricting each agent to its declared capabilities, per-agent and per-tool rate limits, full tool-call audit logging, and a human-in-the-loop approval gateway for high-risk actions — with OpenTelemetry tracing per agent turn and tool call, so agent behaviour is reconstructable after the fact.
- Enterprise Retrieval Architecture:** Designed a dual-path knowledge assistant separating live analytics from indexed content — semantic-model queries executed as the calling user with row-level security passthrough, alongside permission-scoped document retrieval, routed by a query classifier. Metric values are never embedded, so figures cannot go stale; the same multi-tenant pattern scopes access per business unit and country on a client-facing deployment.
- Multi-Agent Operational Intelligence:** Architected cross-domain root-cause analysis over five siloed OSS domains (RAN, Core, Transport, Inventory, Assurance). A query-planning agent decomposes natural-language questions and dispatches to domain-specialist agents holding their own API tools, with a correlation layer synthesising answers cited to source, element, metric and timestamp. Specialist agents over one monolith, because each domain carries its own data model and metric vocabulary.
- Legacy Modernization:** Architected an MCP-based code intelligence system over mainframe COBOL and JCL estates for insurance cloud migration, built on a structure-aware parser and a static dependency graph. The model queries the live codebase through MCP tools rather than an embedded index — chosen because the estate changes as modernization proceeds and answers must cite exact program, paragraph and line, which similarity search cannot guarantee.
- Technical Leadership:** Lead architecture reviews, decision records and trade-off analysis for a team of AI engineers, senior engineers and data scientists; own enterprise AI architecture standards, technical strategy, stakeholder alignment and mentoring.

AI Enterprise Architect

Client: national-scale insurance data ecosystem (India)

Sole architect of the AI Trust Fabric for a national-scale insurance data ecosystem aggregating records from across the country's insurance industry, under national insurance-regulator oversight and data-localisation requirements.

- **Zero Trust Architecture:** Designed the full NIST 800-207 structure — policy administration, decision, enforcement and information points as separate concerns — with enforcement deployed as an API gateway intercept, so every call, human or agent, traverses one boundary rather than relying on per-application checks.
- **Adaptive Trust Engine:** Replaced binary allow/deny with continuous trust scoring across identity confidence, device posture, behavioural baseline, data classification and context, driving allow / step-up-authentication / deny decisions with thresholds configurable per data class. Scores are cached in second-scale windows, trading freshness against decision latency on the hot path.
- **AI Agent Identity & Governance:** Gave agents their own workload identity rather than borrowed user credentials, with action-scope enforcement, per-agent rate limiting, prompt-injection detection, full decision audit, and human escalation above a risk threshold, with trust signals for whether a session's tool-call pattern matches the agent's declared purpose.
- **Offline AI Capability:** Architected fully offline inference with Ollama, and a natural-language policy intelligence layer over insurance regulation orchestrated with LangGraph, to satisfy no-internet-egress environments — a hard constraint that would otherwise have blocked the entire governance layer.
- **Explainability & Audit:** Wired traceable rationale into every access decision — which policy matched, which trust signals fired — against an immutable audit log, making AI-assisted decisions defensible under regulatory audit.

Arcstream Technologies

Aug 2022 – Jan 2025

Technical Security Lead

- **Enterprise Security Modernization:** Led Zero Trust network segmentation and a SASE migration covering 5,000+ users, moving the trust boundary from the network perimeter to identity — the model later extended to AI agents as first-class principals.
- **Secure Delivery & Governance:** Hardened CI/CD pipelines and established architecture governance across engineering teams, embedding security review into the delivery path rather than appending it as a release gate.

Reezanorp Technologies

Dec 2016 – Jul 2022

Senior Software Engineer

- **Applied ML & Data Engineering:** Built machine learning, analytics, computer vision, ETL and enterprise automation systems in Python — the production engineering foundation the later platform and architecture work is built on.

PUBLICATIONS & INTELLECTUAL PROPERTY**MCP Guardian: A Security-First Layer for Safeguarding MCP-Based AI Systems**

Computer Science & Information Technology (CS&IT), Vol.15 No.9, pp.107–121, 2025 · DOI 10.5121/csit.2025.150908 · Co-author
Independent research. Middleware for the Model Context Protocol adding authentication, per-token rate limiting, WAF-style scanning of tool calls and request logging, without modifying individual MCP servers. Evaluated at roughly 3–4 ms median latency overhead.

AI Trust Fabric — Secure Insurance Data Management Device

Registered Design No. 485611-001, Government of India · Co-registrant

The Zero Trust and AI governance architecture designed for a national-scale insurance data ecosystem — NIST 800-207 policy separation, adaptive trust evaluation, identity-centric controls and secure data access mechanisms.

TECHNICAL SKILLS

AI & Agentic Systems: Agentic AI, Multi-Agent Systems, Model Context Protocol (MCP), FastMCP, Agent-to-Agent (A2A), Tool Calling, Human-in-the-Loop, LangGraph, CrewAI, LangChain, Semantic Kernel, AutoGen, Large Language Models (LLM), Generative AI (GenAI), Prompt Engineering

AI Architecture & Governance: Retrieval-Augmented Generation (RAG), Semantic Search, Vector Databases, AI Governance, Agent Identity, Action-Scope Enforcement, Prompt-Injection Defence, Explainability, LLMops, Azure OpenAI, Azure AI Foundry, Ollama

Security: Zero Trust Architecture, NIST 800-207, Policy Administration/Decision/Enforcement/Information Points (PAP/PDP/PEP/PIP), Identity and Access Management (IAM), Microsoft Entra ID, OAuth 2.0, OpenID Connect (OIDC), Multi-Factor Authentication (MFA), Zero Trust Network Access (ZTNA), Secure Access Service Edge (SASE), API Security, Data Loss Prevention (DLP), Row-Level Security (RLS)

Cloud & Platform: Microsoft Azure, Azure API Management, Azure Functions, Azure Logic Apps, Kubernetes (AKS), Docker, Microservices, OpenTelemetry, CI/CD, DevSecOps

Distributed Systems & Data: Event-Driven Architecture, Apache Kafka, Azure Service Bus, Cosmos DB, Microsoft Fabric, Power BI, Azure AI Search, Microsoft Graph, SQL, ETL, Machine Learning

Engineering & Leadership: Python, FastAPI, REST APIs, Legacy Modernization, COBOL/JCL Analysis, Solution Architecture, Enterprise Architecture, Technical Leadership, Mentoring, Stakeholder Management

EDUCATION

Bachelor of Technology (B.Tech), Electronics & Communication Engineering — Jawaharlal Nehru Technological University (JNTU), India